Strategic Investment in a Forward-Looking New Keynesian Monetary–Fiscal Game

Recursive equilibrium, Australian calibration, and parameter regions

Monetary–Fiscal Coordination Project

2 October 2026

The argument in one slide

Mechanism

Adjustment costs make today's instrument a state. An authority can influence its future self and, through the rival's feedback rule, future rival policy.

Forward-looking amplification

Private expectations bring future policy effects into current output and inflation. They amplify an existing rival response; they do not create one.

Quantitative finding

The Australian baseline has policy persistence but weak cross-policy response. Strategic investment is present, yet small.

Robustly large effects require **transmission** $\times$ **durability** $\times$ **rival sensitivity** $\times$ **payoff exposure**.

Research question and contribution

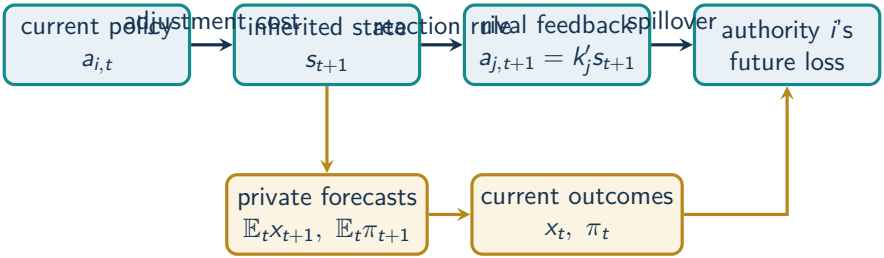
Question

- Can adjustment costs generate endogenous, leader-like positions when monetary and fiscal policy are reset every period?
- How does rational private-sector forward-looking behaviour change that strategic-investment motive?
- Which parameter regions make the motive economically important?

Contribution

- Infinite-horizon monetary–fiscal Markov game.
- Forward-looking IS and Phillips curves.
- Separate agency mandates and true social welfare.
- Rational-expectations fixed point over private forecasts and policy Bellman equations.
- Australian calibration plus strategic-region heat maps.

The strategic-investment mechanism



Crucial distinction

Different objectives create conflict. Strategic investment additionally needs a persistent state and a material rival response to that state.

Forward-looking macroeconomic block

$$
\begin{aligned}
x_t &= \mathbb{E}_t x_{t+1} - \sigma\big(m_t - \mathbb{E}_t \pi_{t+1}\big) + \psi f_t + d_t, \\
\pi_t &= \beta_p \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t, \\
b_{t+1} &= \rho_b b_t + \chi_m m_t + \chi_f f_t - \chi_\pi \pi_t.
\end{aligned}
$$

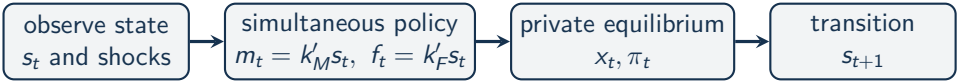
Private sector

Expected future output and the expected real rate affect current demand; expected inflation and the output gap affect current price setting.

Policy state

Debt is intrinsically persistent. Adjustment costs add m_{t-1} and/or f_{t-1} as payoff-relevant states.

Timing and equilibrium concept



- Private agents observe current instruments and know the stationary rules used from $t+1$ onward.
- Each authority optimizes under discretion, taking the rival rule and private forecasting map as given.
- The solution is a **stationary rational-expectations Markov-perfect Nash equilibrium**—not open-loop and not an imposed Stackelberg order.

Agency mandates versus true social welfare

$$J_M = \mathbb{E}_0 \sum_t \beta_M^t \left[\omega_\pi^M \pi_t^2 + \omega_x^M x_t^2 + \omega_b^M b_{t+1}^2 + r_M m_t^2 + \lambda_M \Delta m_t^2 \right],$$

$$J_F = \mathbb{E}_0 \sum_t \beta_F^t \left[\omega_\pi^F \pi_t^2 + \omega_x^F x_t^2 + \omega_b^F b_{t+1}^2 + r_F f_t^2 + \lambda_F \Delta f_t^2 \right],$$

$$J_S = \mathbb{E}_0 \sum_t \beta_S^t \left[\omega_\pi^S \pi_t^2 + \omega_x^S x_t^2 + \omega_b^S b_{t+1}^2 + r_M^S m_t^2 + r_F^S f_t^2 + \lambda_M \Delta m_t^2 + \lambda_F \Delta f_t^2 \right].$$

Delegated objectives

Monetary policy is relatively inflation-oriented; fiscal policy is relatively output-oriented. Each agency bears only its own instrument costs.

Coordination

Coordination minimizes a separate, fixed social loss. It does not average the agency losses or change weights across regimes.

Five institutional experiments

Regime	Adjustment states	Decision maker
Flexible Nash	none	two agencies, simultaneous
Monetary persistence	m_{t-1}	two agencies, simultaneous
Fiscal persistence	f_{t-1}	two agencies, simultaneous
Bilateral persistence	m_{t-1}, f_{t-1}	two agencies, simultaneous
Coordination	none	social planner under discretion

Interpretation

Unilateral persistence gives one authority commitment capacity. Bilateral persistence lets both authorities invest in inherited positions. It is not a literal outer "meta-game" because λ_M and λ_F are primitives.

Private rational-expectations fixed point

Let $y_t = (x_t, \pi_t)'$ and $q_t = (s_t', m_t, f_t)'$. Conditional on a forecast $\mathbb{E}_t y_{t+1} = G s_{t+1}$,

$$
\begin{aligned}
s_{t+1} &= N q_t + E y_t, \\
L y_t &= D q_t + H G s_{t+1}, \\
y_t &= O(G) q_t, \; qquad O(G) = [L - HGE]^{-1}[D + HGN].
\end{aligned}
$$

Partition $O(G) = (O_s, O_m, O_f)$. Equilibrium requires

$$
\boxed{G = O_s(G) + O_m(G) k_M(G)' + O_f(G) k_F(G)'.}
$$

Interpretation

Beliefs determine current private transmission; policy rules respond to that transmission; the resulting outcomes must validate the beliefs.

Policy recursion for a provisional forecast map

The reduced transition is

$$s_{t+1} = A(G)s_t + B_M(G)m_t + B_F(G)f_t.$$

With $V_i(s) = s' P_i s$, simultaneous policy first-order conditions are

$$\begin{bmatrix} Q^M_{mm} + \beta_M B_M' P_M B_M & Q^M_{mf} + \beta_M B_M' P_M B_F \\ Q^F_{fm} + \beta_F B_F' P_F B_M & Q^F_{ff} + \beta_F B_F' P_F B_F \end{bmatrix} \begin{bmatrix} m_t \\ f_t \end{bmatrix} = - \begin{bmatrix} Q^M_{ms} + \beta_M B_M' P_M A \\ Q^F_{fs} + \beta_F B_F' P_F A \end{bmatrix} s_t.$$

Then

$$A_c = A + B_M k_M' + B_F k_F', \qquad P_i = C' Q_i C + \beta_i A_c' P_i A_c.$$

Computation

Inner loop: coupled policy Riccati equations. Outer loop: damped update of G.

What exactly is the strategic-investment effect?

If the rival rule is $a_{j,t+1} = k_j' s_{t+1}$, then

$$\frac{\partial a_{j,t+1}}{\partial a_{i,t}} = k_j' B_i.$$

The continuation FOC can be decomposed schematically as

$$\underbrace{\beta_i B_i' P_i s_{t+1}}_{\text{total continuation effect}} = \underbrace{\mathcal{N}_i}_{\text{own-state effect}} + \underbrace{\beta_i \sum_h B_{ih} k_{jh} \mathcal{H}^i_{j,t+1}}_{\text{rival-response effect}}.$$

Broad dynamic strategy

Rival responses through debt and other macro states.

Petkov channel

Rival response specifically through the inherited own-policy state activated by adjustment cost.

When should strategic investment be large?

- **State exposure**: current policy materially changes a persistent state.
- **Rival sensitivity**: $k_j' B_i$ is steep.
- **Payoff exposure**: induced rival policy matters for authority i.
- **Patience**: future losses receive substantial weight.
- **Expectations transmission**: $\|[L - HGE]^{-1} H\|$ is large.
- **Feasible movement**: persistence is strong enough to bind the future, but not so strong that moving today is prohibitive.
- **Stability**: amplification does not merely reflect proximity to a unit root.

Response-product logic

These ingredients are complementary. Mandate disagreement alone is not enough.

Australian quarterly calibration

Structural parameter	Value
σ	0.03695
ψ	0.05142
κ	0.15511
β_p	0.99
ρ_b	0.995
$(\chi_m, \chi_f, \chi_\pi)$	$(.015, .060, .100)$
(λ_M, λ_F)	$(.05, .04)$

Weights	Inflation	Output	Debt
Monetary	1.50	0.25	0.15
Fiscal	0.50	0.75	0.15
Social	1.00	0.50	0.15

Interpretation

Transmission coefficients are descriptive Australian estimates, not a fully identified structural NK estimation. Weights and adjustment costs are transparent calibrations.

Data inputs: Reserve Bank of Australia statistical tables.

Impact policy responses

Demand contraction

Regime	m_0	f_0	Actual loss
Flexible Nash	-0.0874	-0.0843	0.1154
M persistence	-0.0660	-0.0802	0.1153
F persistence	-0.0883	-0.0648	0.1154
Bilateral	-0.0667	-0.0615	0.1153
Coordination	-0.1369	-0.0540	0.1149

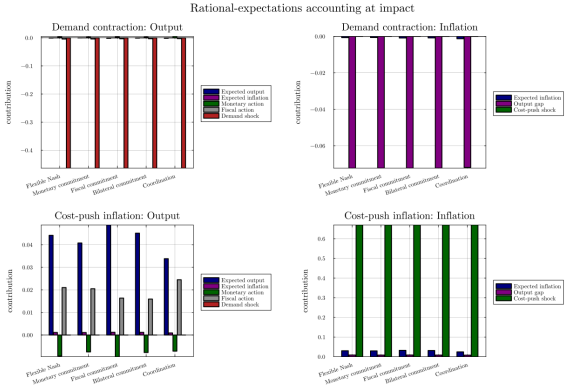
Cost-push inflation

Regime	m_0	f_0	Actual loss
Flexible Nash	0.2516	0.4091	0.5428
M persistence	0.2029	0.3989	0.5413
F persistence	0.2606	0.3182	0.5476
Bilateral	0.2098	0.3099	0.5460
Coordination	0.1934	0.4763	0.5325

Why can coordinated monetary policy move more?

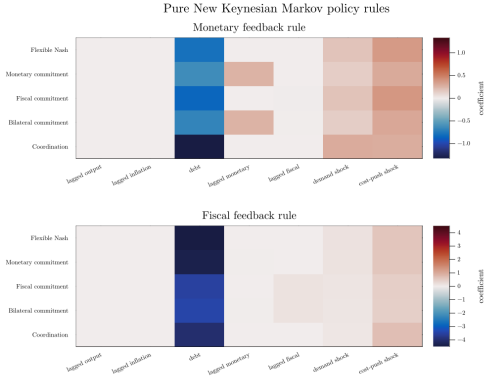
The planner internalizes both instruments' spillovers and reallocates the joint response under fixed social weights. The result is not driven by changing the objective between regimes.

How expectations enter on impact



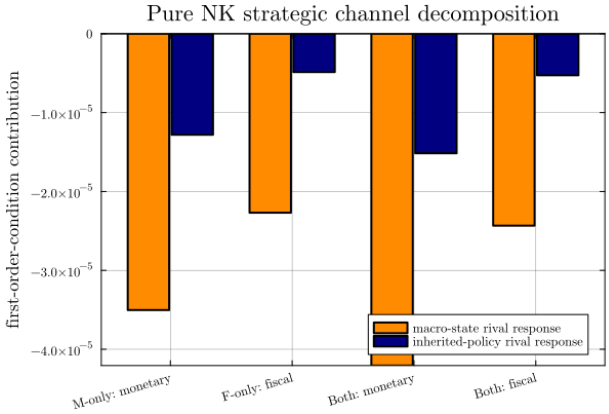
- The transitory demand innovation dominates impact output.
- Following a cost-push shock, expected future output contributes visibly to current activity.

Policy rules: persistence is not yet leadership



- Own lag coefficients: roughly 0.26 for monetary policy and 0.17 for fiscal policy when the corresponding adjustment cost is active.
- Cross-lag coefficients are much smaller: the future self is bound more strongly than the rival is moved.

Baseline strategic channel decomposition



Demand shock

Inherited-policy bounded share:

- monetary persistence: 0.003;
- fiscal persistence: 0.002.

Diagnosis

The model creates commitment capacity, but the rival reacts only weakly to the inherited policy position.

Parameter-region experiment I

Axes

- Policy-transmission scale τ multiplies σ and ψ.
- Dimensionless persistence share

$$p_i = \frac{\lambda_i}{D_i + \lambda_i}.$$

- Australian values: $p_M = 0.330$, $p_F = 0.282$.

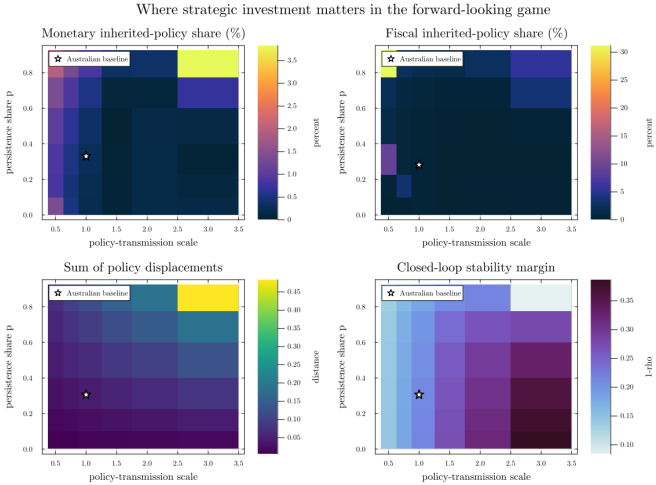
Diagnostics

- inherited-policy share of the continuation FOC;
- policy displacement from flexible Nash;
- closed-loop stability margin $1 - \rho(A_c)$.

Reading rule

A high share is persuasive only when policy displacement is material and the equilibrium remains comfortably stable.

Where strategic investment matters



Interpreting the transmission–persistence map

Robust high-effect corner

At $\tau = 3.00$ and $p = 0.85$:

- policy displacement $= 0.484$;
- stability margin $= 0.084$;
- monetary inherited share $= 3.83\%$.

Strong transmission and durability reinforce one another, but stability thins.

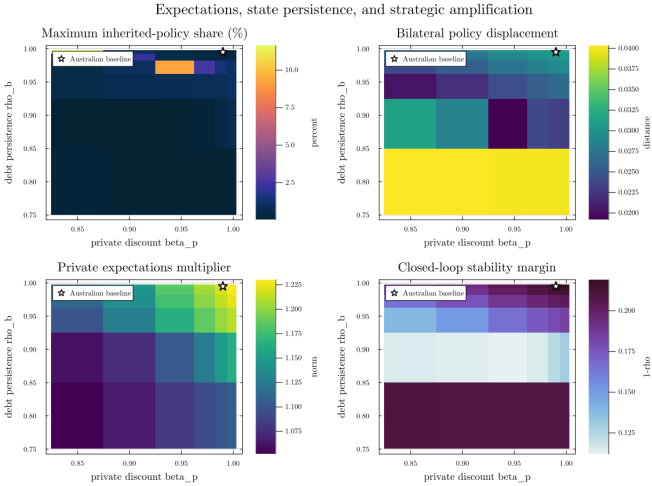
A high-share false positive

Fiscal share reaches 31.19% at weak transmission and $p = 0.85$, but:

- displacement is only 0.071;
- the large share reflects cancellation in a small denominator.

All 36 grid equilibria converge and are stable.

Expectations and debt persistence



Forward-lookingness is an enabling condition

What rises monotonically

The private expectations multiplier grows with β_p and reaches 1.230 in the grid.

What does not

The inherited-policy strategic share is non-monotonic in β_p and ρ_b.

Why

Stronger expectations can both transmit future rival policy and improve anticipated stabilization, reducing the shadow value of manipulation. Persistent debt may make both agencies react to debt rather than to one another's inherited instrument.

Expectations amplify $k'_j B_i$; they do not substitute for it.

Equilibrium selection and fiscal dominance

Equilibrium search

- 7 dispersed initial forecasting maps.
- 1 distinct stable stationary solution.
- Bilateral spectral radius: 0.7834.
- All 72 parameter-map cells converge and are stable.

No baseline evidence of difficult equilibrium selection—not a global uniqueness theorem.

Fiscal dominance boundary

The model has fiscal pressure through debt and inflation, but does not yet include nominal government valuation or alternative monetary/fiscal backing regimes. It is not a complete fiscal-theory-of-the-price-level model.

Conclusions

1. Adjustment costs create a genuine strategic state and partial commitment capacity.
2. Forward-looking private behaviour transmits future policy into current output and inflation.
3. Different mandates create conflict, but large strategic investment additionally requires a strong rival response.
4. The Australian baseline binds each authority's future self more than it moves the rival.
5. Robustly large effects arise where strong transmission and high persistence coincide—often with a thinner stability margin.

Next empirical target

Estimate the cross-policy feedback slopes and the full response product, then add persistent shocks and nominal government liabilities.

Selected references

- Clarida, Galí and Gertler (1999), "The Science of Monetary Policy."
- Beetsma and Bovenberg (1997), "Central Bank Independence and Public Debt Policy."
- Jun and Vives (2004), "Strategic Incentives in Dynamic Duopoly."
- Petkov (2004), "Delegated Management in Dynamic Oligopolies."
- Muscatelli, Tirelli and Trecroci (2004), "Fiscal and Monetary Policy Interactions."
- Kirsanova and Yates (2025), "Multiple Equilibria in the Absence of Commitment."

Questions and discussion

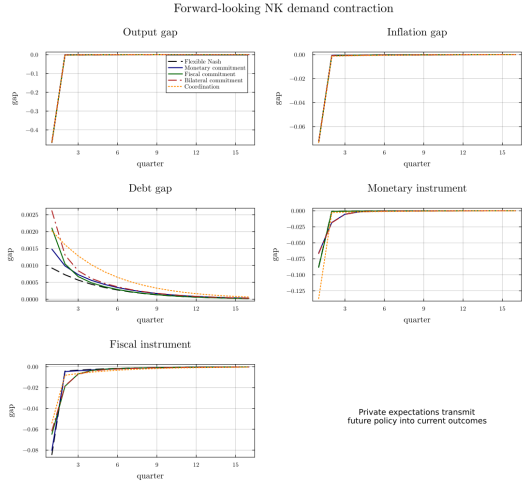
Backup: numerical algorithm

1. Initialize private forecasting matrix G.
2. Construct $O(G)$ and reduced transition matrices A, B_M, B_F.
3. Solve the coupled policy Riccati equations for k_M, k_F, P_M, P_F.
4. Form the implied private forecasting map $\Phi(G)$.
5. Update $G \leftarrow (1-\delta)G + \delta\Phi(G)$ and repeat.
6. Verify private equations, FOCs, Bellman residuals, and $\rho(A_c) < 1$.

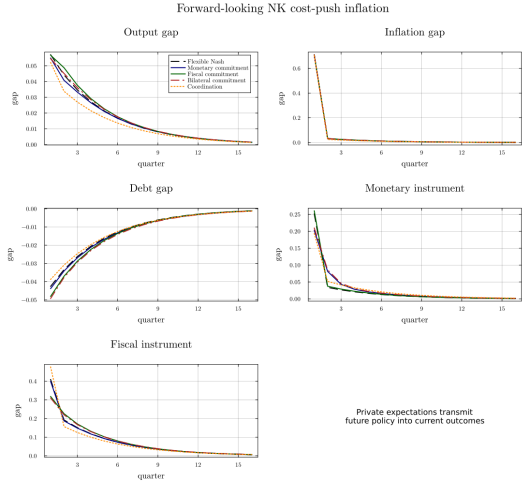
Baseline computation

The bilateral equilibrium takes 278 damped outer iterations. The Julia test suite contains 52 passing checks.

Backup: demand-shock impulse responses



Backup: cost-push impulse responses



Private expectations transmit future policy into current outcomes

Backup: welfare accounting

- **Macro loss**: common social inflation, output, debt, and instrument-level terms; excludes adjustment costs.
- **Actual loss**: adds the real implementation cost of adjustment.

Demand experiment	Macro loss	Actual loss
Flexible Nash	0.1154	0.1154
Monetary persistence	0.1149	0.1153
Fiscal persistence	0.1151	0.1154
Bilateral persistence	0.1147	0.1153
Coordination	0.1149	0.1149

Interpretation

Inertia can improve macro stabilization while still consuming real implementation resources. Coordination and commitment capacity are distinct.

Backup: how to extend toward fiscal dominance

- Replace the reduced-form debt law with a consolidated nominal government budget constraint.
- Introduce nominal maturity structure and the market value of debt.
- Specify monetary versus fiscal backing regimes.
- Let the price level participate in government liability valuation.
- Re-solve the policy game to ask whether strategic investment shifts the boundary between active monetary and active fiscal regimes.